Measurements and analysis of electrical power consumption patterns for the computation of an aggregated battery energy storage system

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***Abstract*—Aggregated and distributed battery energy storage systems improve distribution grids operability by providing advanced energy management schemes and energy resources allocation. Charging over consumer's peak power is one of the implementations of energy contracts that incentivize storage, even if it does not provide a direct interconnection between intermittent renewables and storage increasing its benefits. In this paper, an aggregated battery energy storage system is computed based on measured electrical power consumption patterns of a residential distribution network. Aiming to estimate the techno-economic factors that could potentially incentivize the installation of the battery system, the consumption patterns at both the transformer and the load side are analysed. The parameters that affect the economic viability of the system are evaluated in order to perform financial calculations about the profitability of the system.**

Keywords—batteries, consumption patterns, distribution network, energy storage, peak-shaving

I. Introduction

Future energy systems show strong de-centralized characteristics in the presence of renewable energy sources (RES), aggregated or distributed energy storage systems (ESS) and flexible loads such as electric vehicles (EVs). Due to new open market business models, many customers have the opportunity to produce and store energy and consequently, they have an active role in the electrical power grid. This alters the distribution grid from a passive grid to an active grid with bi-directional power flow, the customers become prosumers (producers-consumers) and in this manner, the conventional centralized power systems evolve strong de-centralized characteristics [1, 2]. Peak-shaving using storage has been previously investigated without being always connected to the applicable energy contracts. S. Favuzza et al. [3] at project DEMAND created an algorithm that aims to completely flatten consumer's load profile. This is a valid approach, nevertheless, overacting could require massive storage capability that could be techno-economically unfeasible to be operated. It is therefore important to understand using real data and information the applicability of any relevant proposal. Jaser Sa'ed et al. [4] present an optimization methodology that provides an holistic approach for dimensioning a battery energy storage system based on the applicable peak charging energy contracts. Bwo-Ren Ke et al. [5] provide a procedure for sizing an energy storage system that operates in line with a photovoltaic system. This approach clearly provides benefits as far as the energy bills are concerned, but these benefits are rather produced from the solar system than the battery storage per se. Enrique Lobato et al. [6] develop an algorithm for peak shaving using energy storage for isolated networks. It is applied to Spanish Canary islands with adequate results, nevertheless, the operational characteristics of a storage device at the interconnected system shall be treated in a different manner. Roozbeh Karandeh et al. [7] investigate peak shaving through energy storage for charging applications. They demonstrate the impact of the storage to the capability of the complete system to provide smooth load profiles even under intense operation periods.

In this paper, the electrical power consumption patterns of a real residential distribution network were studied in order to develop and compute an aggregated battery energy storage system (BESS). The measurements at both the transformer and the load side are analysed aiming to evaluate the techno-economic factors that potentially incentivize the installation of aggregated BESS. Based on the measurements, an aggregated BESS at the transformer side is computed in order to evaluate the parameters that affect the BESS economic viability as well as financial calculations for the profitability are performed.

II. System Description and Measurements

Customers utilize energy in different consumption patterns according to their daily activities and habits. To create an understanding of these variations for a holistic view of how to develop an aggregated BESS as well as how the simulation models should be programmed, in the next sections electrical power consumption patterns at the transformer side and at the load side were studied.

A. Description of the System

The part of the residential network in Fig. 1 at Karlstad city in Sweden is examined in this paper. An 800kVA distribution transformer supplies 69 customers from which 68 consist of typical residencies and 1 represents a collection rail for lamp posts. The regional distribution system operator (DSO) provided access to the meters of the network in order to collect measured data of the consumed power during one year, from 1st March 2018 to 28 February 2019. The measurements have been collected and processed accordingly

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in the form of lists. For sensitivity reasons the measurement are presented in the form of graphs in the next section.

B. Measured Data at Transformer Side

The collected data at the transformer side during one year are shown in the Fig. 2. In Fig. 2(a), one can observe a decreased energy consumption during spring and summer in comparison to autumn and winter seasons. This is due to the higher temperature about 15ºC - 25ºC when there is no special need for space cooling. The power peak varies between 80-290kW depending on the season. Moreover, one can observe that during May to mid-September there are no high monthly variations of the power consumption as those in comparison to the period between mid-September to end of April. For example, during November the lowest and the highest peaks are about 75kW and about 240kW, respectively while June they are about 25kW and 80kW, respectively. Fig. 2(b) shows the consumption from June to August during weekdays. By following the blue line that shows the average consumption, there are two high peaks, one in the morning around 08.00 having value about 60kW and one in the evening around 18.00 having value about 75kW.

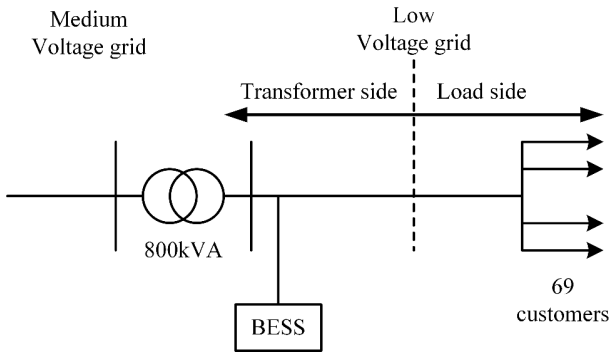

Fig. 1. The one-line diagram of the residential distribution part of the network under examination and placement of the BESS.

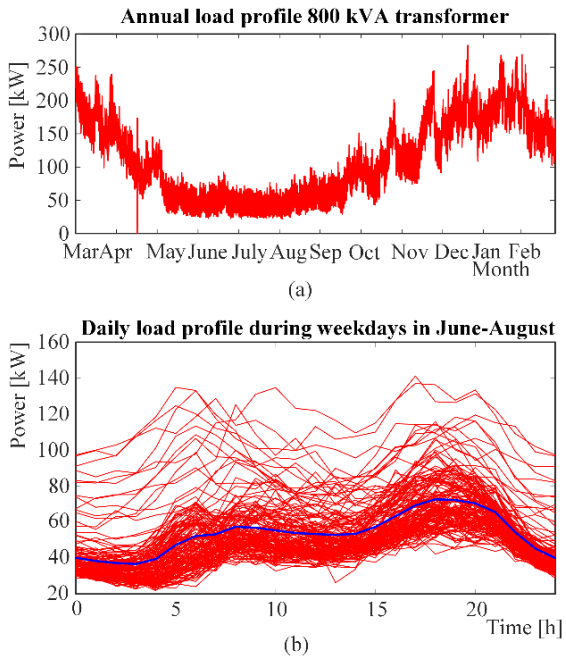

Fig. 2. Power consumption (a) from 1st March 2018 to 28 February 2019 and (b) during weekdays from June to August 2018.

Fig. 3 presents the consumption patterns for the period between November to March during weekdays and weekends. During weekdays and by reading the blue average line in Fig. 3(a), one can observe two distinct highest peaks. The first is in the morning around 07.00 having a value about 175kW and the second is in the evening around 20.00 having value about 190kW. During weekends in Fig. 3(b), by reading the average blue line, one can observe that the highest peaks reach the same levels as those during the weekdays, the evening highest peak is around 20.00 however, the highest peak in the morning is shifted around 10.00 in comparison to that in weekdays.

The above analysis of the measured consumption patterns reveal that except of the high seasonal variation as is presented in Fig. 2(a), the consumption patterns from the transformer side reveal strong similarities during particular seasons. As such, the transformer does not experience loading conditions that have high differences and consequently, a technically and economically viable aggregated BESS system for energy storage and peak shaving could be computed suitable to cover all customers treated as one aggregated load.

C. Measured Data at the Load Side

The individual consumption patterns for each customer have been measured. Fig. 4 presents the power peak and the total power consumption of the residential area under examination. The peak power varies between 3.6kW to 12.7kW while the annual energy consumption varies from 9MWh to 26.3MWh.

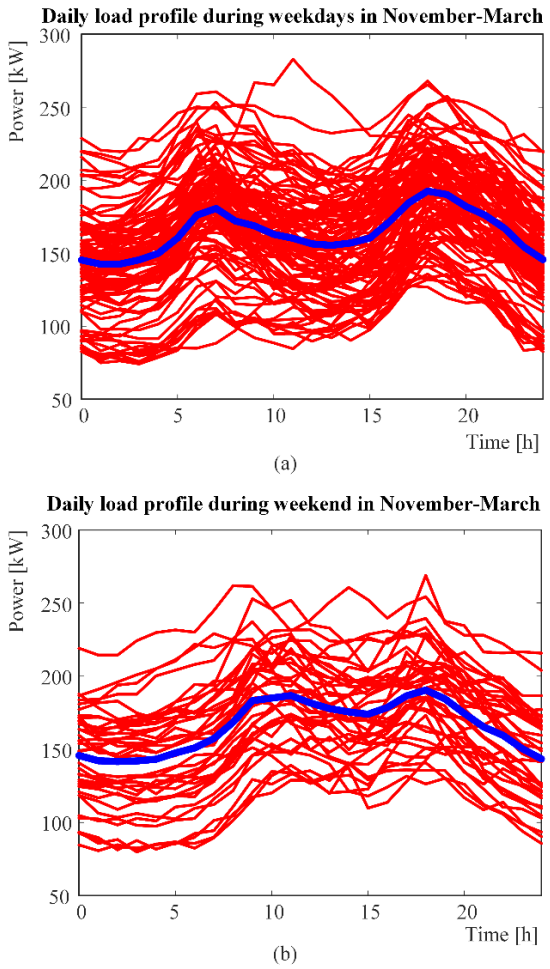

Fig. 3. Power consumption from 1st November 2018 to 31rd March 2019 (a) during weekdays and (b) during weekends.

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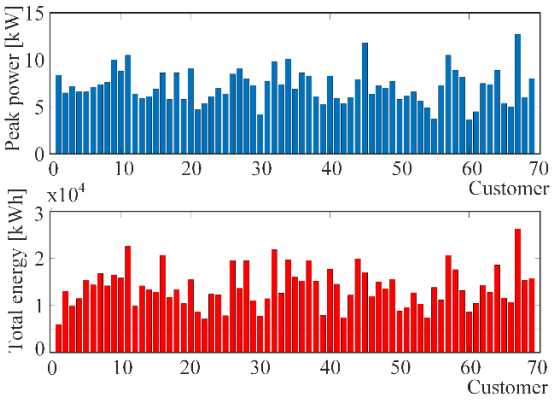

Fig. 4. The power peak and the total power consumption of the residential area under examination for each customer from 1st March 2018 to 28 February 2019.

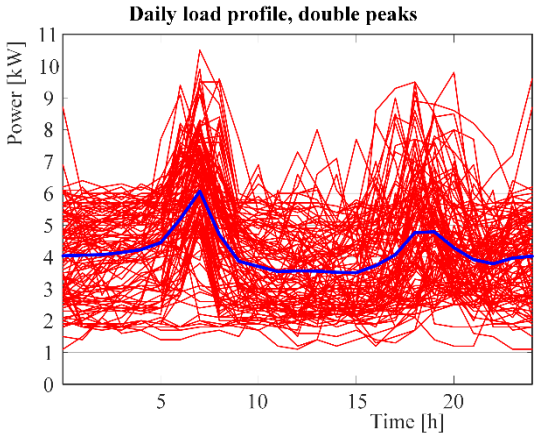

Fig. 5. The double-peaks consumption pattern.

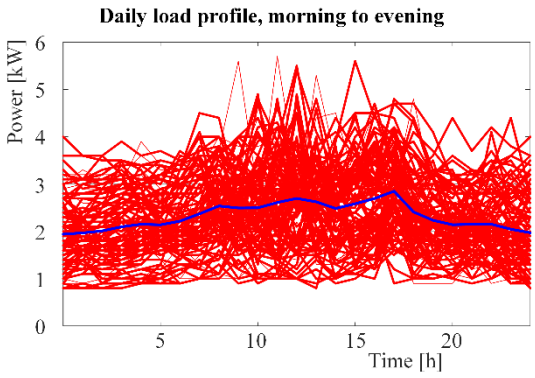

Fig. 6. The day-evening consumption pattern.

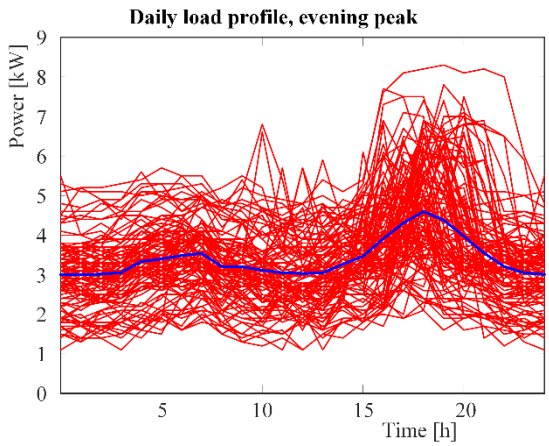

Fig. 7. The evening peak consumption pattern.

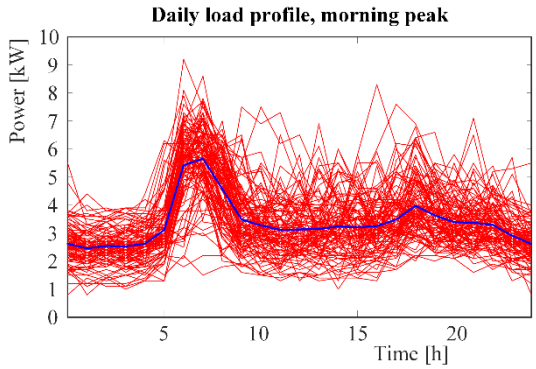

Fig. 8. The morning peak consumption pattern.

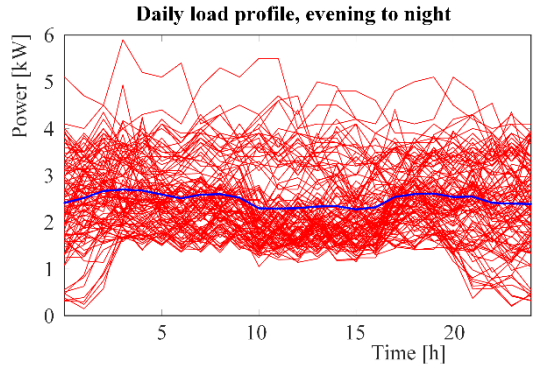

Fig. 9. The evening-night peak consumption pattern.

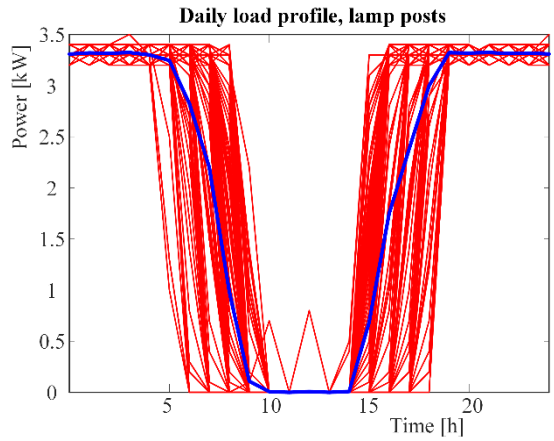

Fig. 10. The lamp posts consumption pattern.

As the highest power output occurs during the winter months, the residential consumption patterns were further analysed between November and March because during these months there are additional fees from the regional DSO called as “high-charge fees”. The analysis reveal six consumption patterns. There are five different residential consumption patterns during weekdays and are presented in the Fig.5 to the Fig. 9. They are categorized as double-peaks, day-evening, evening-peak, morning-peak and evening-night. There is also one pattern that corresponds to outlets of lamp posts in the Fig. 10. Among the above defined profiles of the 69 consumers, the 45% presented double-peaks and 39.1% presented day-evening peaks. The rest presented 8.7% evening-peak, 4.3% morning-peak and 1.45% evening-night. The final 1.45% corresponds to the lamp posts.

The measured residential consumption patterns reveal that the patterns may appear major differences. Residential storage and peak-shaving, especially in the case where the customer

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operates as prosumer, is case-by-case depended. Hence, none may take as given that energy storage should be conducted during night supposing that the energy demands are lower. Analysis of the individual consumption pattern for each prosumer is necessary in order to size an economically viable residential BESS. Another important finding from the analysis of the measurements is that the customers which may bring energy production from RES and residential BESS may participate in peer-to-peer (P2P) energy trading because their peak demands rise on different time during a 24 hours period. This kind of energy management schemes for optimal trading based on market tariffs is considered an important topic which attracts high attention in the literature that however, is not addressed in this particular investigation.

III. Computations on the Aggregated BESS

The aggregated BESS computations are performed by using the flowchart, the equations and the conditions presented in the appendix. In this section, simulation results such as the state of charge (SOC), the depth of discharge (DOD) and the C-rate as well as discussions and challenges regarding financial calculations are presented. Preliminary investigation has been conducted in order to define the threshold limit and the battery size. The BESS under development aims to sustain the transformer during the peak periods from beginning of December to end of March rather than during the whole year. In addition, a lower threshold limit leads to larger battery size and thus, to a more expensive investment. As such, the presented investigation is focused in the period beginning of December to end of March.

A. Simulation Results of the BESS

Fig. 11 shows the reduction of the power peak at the transformer side. The maximum power output is 282.8 kW, the threshold limit of 246 kW and capacity of the battery is 289 kWh. This leads in a reduction of about 13% (36.8 kW). The simulation settings yielded a maximum charge rate and discharge rate of 21.32 kWh and 29.49 kWh respectively shown in Fig. 12. This corresponds to 0.074C and 0.102C for the C-rate, respectively. Battery discharge occurs only for 13 days throughout the year from December to March leading to DOD variation between 33%-70%.

Further simulation have been conducted by lowering the threshold limit and increasing the capacity of battery. The results present qualitative similarities to those presented in Fig. 11 and in Fig. 12 such as higher reduction levels, C-rates and more days of discharge. However, these results should be investigated by an economic perspective in order to estimate their influence on the total investment. This investigation is presented in the next section.

B. Economic Calculations

In this section, total profit is calculated for the above computed aggregated BESS at the transformer side for increasing threshold limit from 246 kW onwards. Table I shows the savings per year, the repayment period and total profit depending on the threshold and the battery price. Savings per year are calculated as the difference of the charges per year before and after reduction by using tariffs as 194SEK per kW and year for the power charges as well as 0.007SEK per kWh for varaying regional network charges as they have been taken from the regional transmission operator. Larger reduction results in lower total profit because of the battery size and cost.

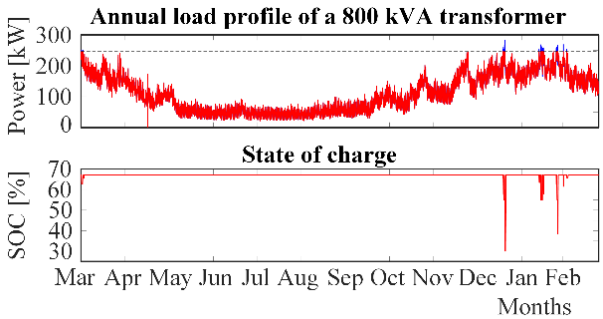

Fig. 11. Peak-shaving at the transformer side.

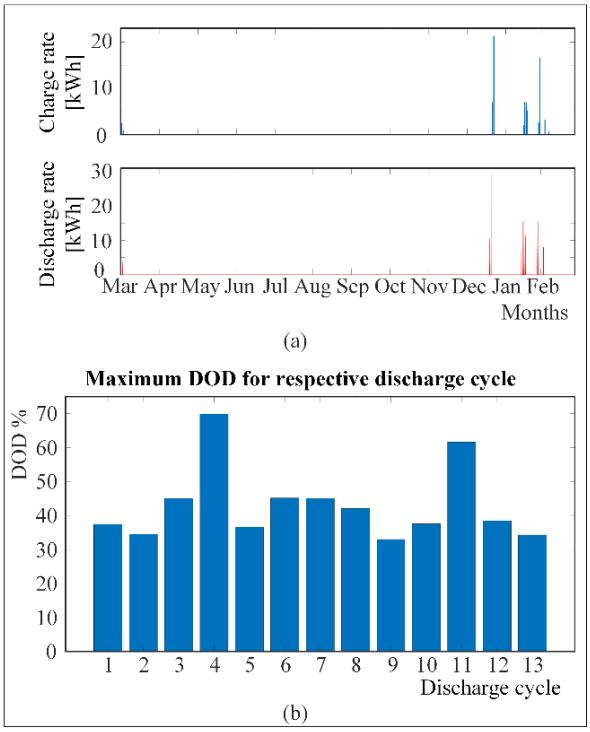

Fig. 12. The charge and discharge rate and the DOD.

TABLE I. Economic Calculations Before and After Reduction

Reduction kW	0	17.8	27.8	36.8
Battery Capacity [kWh]	-	57	166	289
Power charge/year [SEK]	54683	51410	49470	47724
Variable charge/year [SEK]	6660	6674	6695	6721
Savings/year [SEK]	-	3439	5358	7077
Payback for 100$/kWh [years]	-	15.89	29.71	39.55
Payback for 50$/kWh [years]	-	7.94	14.87	19.56
Total profit for 50$/kWh [SEK]	-	41474	27487	3114

Fig. 13 highlights that the annual peak power charge decreases linearly with reduced power, while the variable electricity grid charge stays at approximately the same level between 0-13.9 kW reduced power. Thereafter, the electricity grid variable fee rises almost linearly in relation to reduced power. Moreover, after 13.9 kW reduction of power peak, the total power output increases linearly in relation to reduced power.

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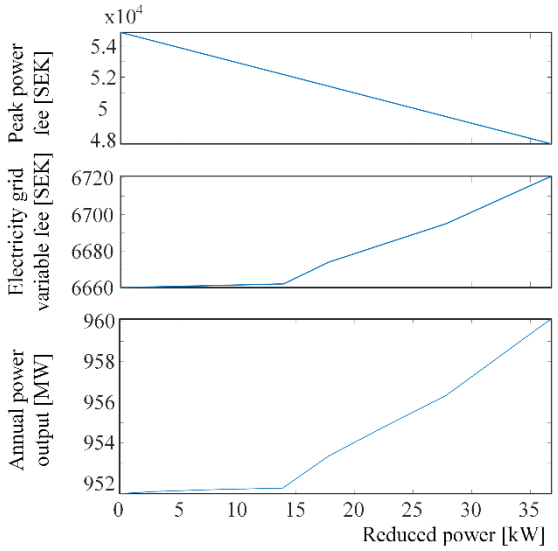

Fig. 13. The peak power charge, the variable grid charges and the annual power output in respect to the reduced power.

The savings per year S_p in respect to the profit time T for battery price 50$/kWh is plotted in Fig. 14. Profit time is the time left after the payback time and the battery's lifetime, which selected to be 20 years. MATLAB curve fitting toolbox has been used in order to compute the curve that best fits to the plotted points. The result is the linear expression

$$S_p = a \cdot T + b \tag{1}$$

$a = -312.2$ SEK/year and b = 7121 SEK.

Total profit Pr is given by

$$Pr = T \cdot S_p \tag{2}$$

By substitution (2) in (1), the total profit Pr shows a second order dependence from the profit time T as

$$Pr = a \cdot T^2 + b \cdot T \tag{3}$$

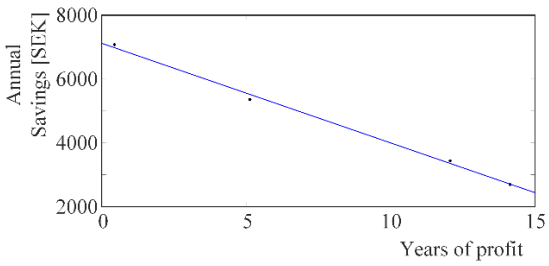

Fig. 14. Saving per year in respect to the years of profit supposing 50$/kWh battery price.

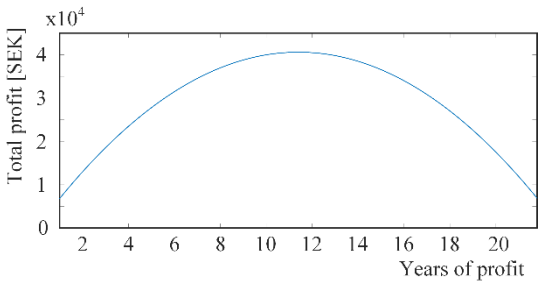

Fig. 15. The total profit versus the years of profit.

Fig. 15 shows that the profit period is 11.4 years at the maximum of the function. The profit period of 11.4 years corresponds to 3560 SEK in savings per year and the payback period of 8.6 years. Total profit is estimated at 40584 SEK that means a reduction of 17.8 kW and battery capacity of 57 kWh. The reason why a reduction of 13.9 kW does not provide maximum profit is because the relationship between savings per year and the profit period is not completely linear.

IV. Conclusions

Electrical power consumption patterns were studied in order to develop and compute an aggregated BESS for a real residential distribution network. Measurements at both the transformer and the load side were performed at 15´ time intervals. Linear interpolation is used at times instants in between the measuring points and this might be a source of error for the most accurate determination of the highest peaks.

At the transformer side, the power output is highest at 07:00 and 17:00 and the lowest between 22:00 - 05:00 on weekdays. On weekends, the power output is at its peak at 10:00 and 18:00 and the lowest between 00:00 - 07:00. The measured consumption patterns reveal high seasonal variation however, strong similarities during particular seasons revealed as well. Consequently, technically and economically a viable aggregated BESS for energy storage and peak shaving is computed suitable to cover all customers treated as one aggregated load.

At the load side, the measured residential consumption patterns revealed major differences leading to six consumption patterns. Hence, residential storage and peak-shaving, especially in the case where the customer operates as prosumer, is case-by-case depended. It is not wise to take as given that energy storage should be conducted during night supposing that the energy demands are lighter. Thus, analysis of the individual consumption pattern, especially when the customer brings RES capabilities, is necessary in order to size an economically viable residential BES. It should be also highlighted that the customers which bring RES and BESS may participate in P2P energy trading because their peak demands rise on different time during a 24 hours period [8]. Optimal trading based on market tariffs is considered an important topic, which attracts high attention in the literature that however, was not addressed in this investigation.

An aggregated BESS was computed by using simulation models built in MATLAB. In the simulation models, the system loss is set to 15% and SOC may vary between 30%-75% to protect battery health. The threshold limit and battery capacity are set to 246 kW and 289 kWh respectively, which corresponds to a 13% reduction. The charge C-rate is 0.074 while the discharge C-rate is 0.1. Discharge takes place only 13 days a year between December-March with DOD varying between 22%-70%. The battery can therefore be used for other applications such as a special AC or DC power lines for electrification of the wider transport sector and EVs integration [9]. At present, the eligibility to install and utilize BESS is under investigation according to national and European laws. Several concepts are under examination according to which, BESS might be integrated either by separate energy storage companies or by individual customers. The customers will participate under a predefined contract having a predefined percentage of ownership of the storage investment.

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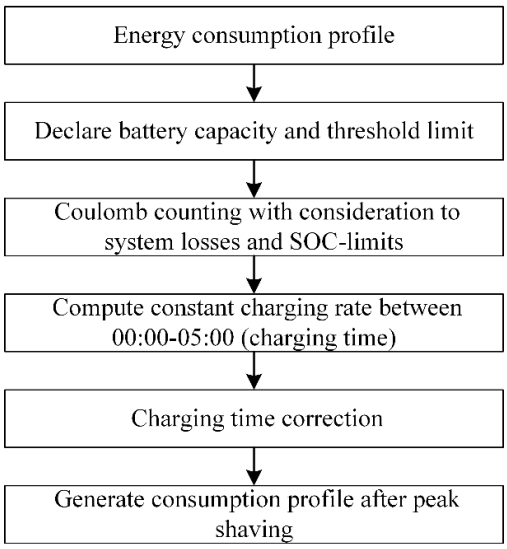

Fig. 16. The simplified block-diagram for the calculations.

Financial calculations for the BESS show that profitability depends on the threshold limit, battery capacity, payback time, profit time and total profit. The calculations also show that the optimal threshold limit does not provide the highest profitability due to a high battery capacity, which results to increased battery cost. The total profit in relation to the profit period appears a second-order dependency. For this particular case study at the transformer side, the total profit is at most 41474 SEK. Investment of a battery system is recommended from 2030 onwards due to drastically falling battery prices in the coming years.

Ongoing work is conducted to develop a distributed at the load side BESS. Thereafter, the comparison between the aggregated and the distributed BESS will reveal the advantages and the disadvantages in each application. Moreover, new simulation models will be developed that take into account the integration of EVs and RES for even more realistic future studies.

APPENDIX

The Coulomb method estimates the SOC and is based on integration of the battery current time-dependent function when charging / discharging the battery [10] and [11]. By considering that the energy consumption is measured in kilowatt hours, SOC is calculated as

$$SOC(t) = SOC(t_0) - (100/C_{kWh}) \cdot \int P(t)dt \qquad (4)$$

where C_{kWh} is the nominal battery capacity in kWh and $P(t)$ is power as a function of time. The DOD is calculated as

$$DOD(t) = 100 - SOC(t) \qquad (5)$$

A simplified block-diagram of the calculations is shown in Fig. 16. The calculation start by reading customers load profile and declaration of battery capacity and threshold limit. Thereafter Coulomb counting method with consideration to system losses 15% is implemented in order to calculate SOC which varies between 75% - 35%. Constant charging rate is executed between 00:00 - 05:00. If the power output is equal or exceeds the threshold limit during charging time, the charging time is postponed by one hour. Load profile after peak shaving can be generated by accumulating the original load profile with charging rate during charging time. The load profile is set to be equal to the threshold limit if the power output is greater than the peak value limit.

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